

In the Age of AI, What Should Public Health Researchers Do?

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For readers of this piece, I assume you already have a solid understanding of AI's capabilities and developmental trajectory, and that you have been exposed to extensive discussions on the topic. There are far too many issues surrounding AI to address; here, I will focus solely on the question posed in the title.

In scientific research, what roles should AI and humans play? Given AI's efficiency and increasingly human-like performance, we can no longer regard it merely as a tool. In research practice, it should be considered a member of the team—indeed, even the primary executor. However, reconstructing roles does not imply human withdrawal. Rather, it implies an upward shift in responsibility: humans must serve as the intellectual leaders, critical reviewers, and team managers—determining research direction, making final decisions, and enabling the “human–AI research team” to maximize its collective intelligence.

To set research direction appropriately, we must possess high-level cognition, deep disciplinary understanding, and a broad strategic perspective in public health. AI's efficiency allows research to proceed rapidly and smoothly, enabling swift responses to emerging public health threats and thereby strengthening public health action. Yet genuine public health capacity should not be limited to passive response, we must preserve our capacity for independent thinking—especially the ability to perceive subtle signals and infer larger trends. Public health problems should not have to appear in surveillance datasets before they are recognized and addressed—researchers must be able to detect latent risks from everyday social phenomena. For example, when you know someone is hesitant about getting vaccinated, this should prompt awareness of potential distrust in the healthcare system. If we wait until vaccination coverage declines in official statistics before recognizing the risk, the response is already delayed. Early recognition enables early intervention and containment of emerging threats.

During research, we must critically evaluate AI-generated analyses. This point is self-evident and requires no extended discussion.

Because AI has become an indispensable member of the research team, project leaders must understand it as thoroughly as they understand human collaborators—its strengths, its limitations, and the conditions under which it performs optimally—so that its potential can be fully realized within collaborative workflows.

Moreover, public health problem-solving inherently requires multidisciplinary thinking and collaboration. Historically, achieving deep expertise in one field required years of specialized accumulation; crossing into another complex discipline often exceeded an individual researcher's cognitive and temporal capacity. With AI, however, researchers can rapidly acquire structured knowledge in unfamiliar domains and conduct sophisticated analytical tasks. AI can thus facilitate collaboration across disciplines, enabling a more comprehensive understanding of

public health challenges. This evolution implies that beyond specialists with deep disciplinary mastery, we also require scholars with intellectual breadth—individuals capable of guiding multidisciplinary research agendas.

Through such approaches, the “human–AI research team” may achieve substantial progress at the level of theoretical analysis. However, public health challenges cannot remain confined to theory. The article *Unintended consequences of the ‘bushmeat ban’ in West Africa during the 2013–2016 Ebola virus disease epidemic* by Bonwitt et al. powerfully demonstrates this point: policies that lack understanding of social realities may generate severe unintended consequences. In an era jointly shaped by AI and humans, public health researchers must compensate for what AI lacks—namely, practical wisdom. Such wisdom may well require engagement with philosophical traditions.

For example, the Northern Song philosopher Xie Liangzuo, summarizing Confucian thought, wrote: “The sage speaks of the ordinary and not the anomalous; of virtue and not of coercive force; of governance and not of problems; of realities and not of abstractions. (圣人语常而不语怪, 语德而不语力, 语治而不语乱, 语人而不语神。)”

This framework holds practical implications for public health:

- **Speak of the ordinary, not the anomalous:** prioritize common conditions and widespread risks rather than allocating attention to rare anomalies.
- **Speak of virtue, not of coercive force:** emphasize education and trust-building over coercive governance.
- **Speak of governance, not of problems:** focus on pathways to resolution rather than merely identifying problems.
- **Speak of realities, not of abstractions:** attend to real social structures rather than abstract theories or slogans.